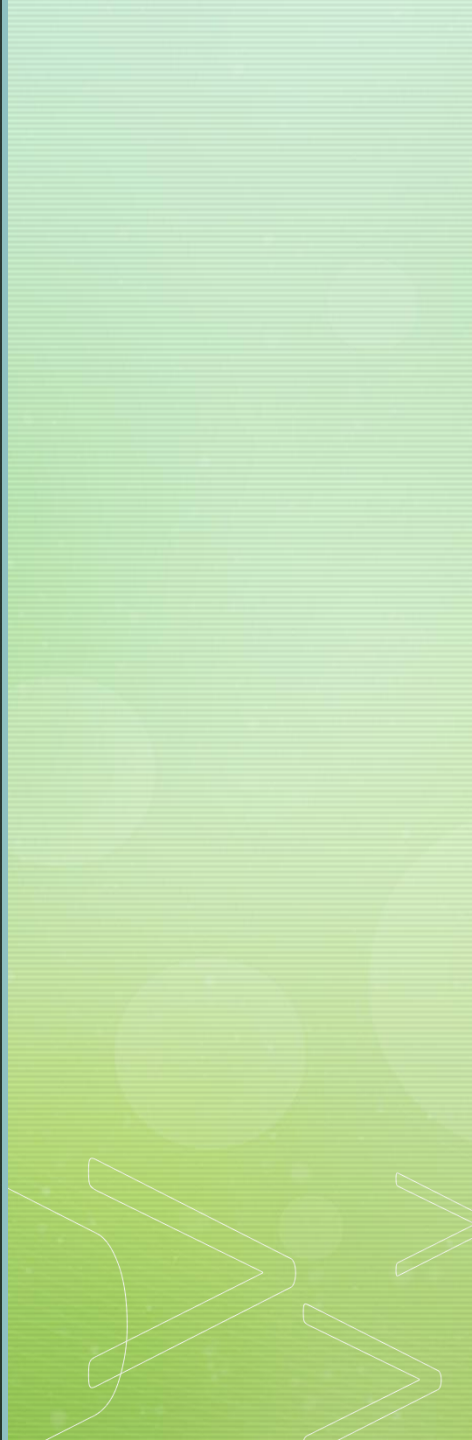


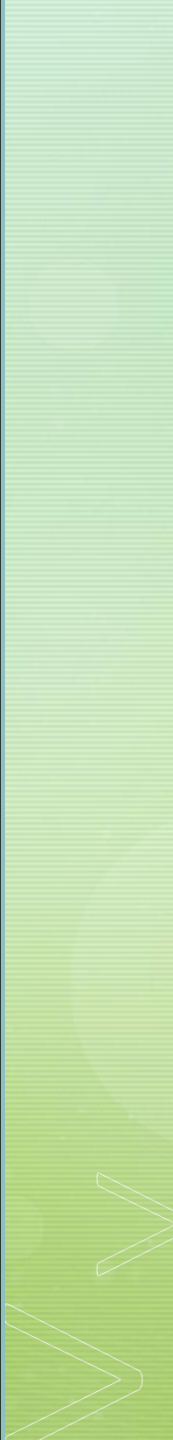


Ethics in Digital Forensics



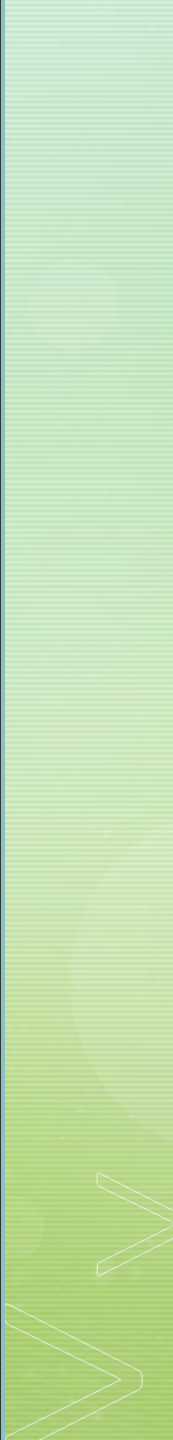


What is Digital Forensics?

- Definition: Investigating digital devices to collect evidence
 - Applications: Cybercrime, corporate security, legal cases
- 

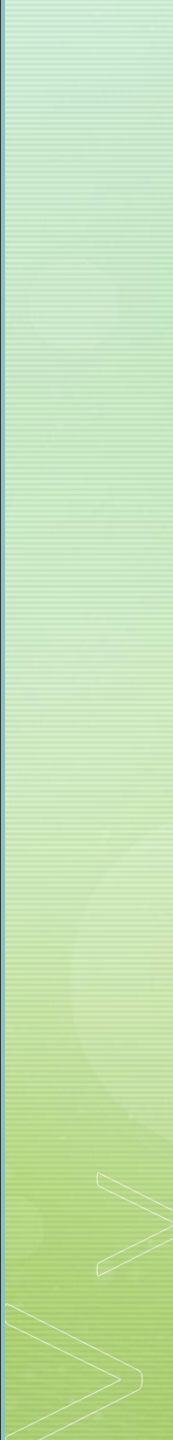


Why Ethics Matters

- Sensitive personal data involved
 - Risk of privacy violations
 - Maintaining legal admissibility and fairness
- 

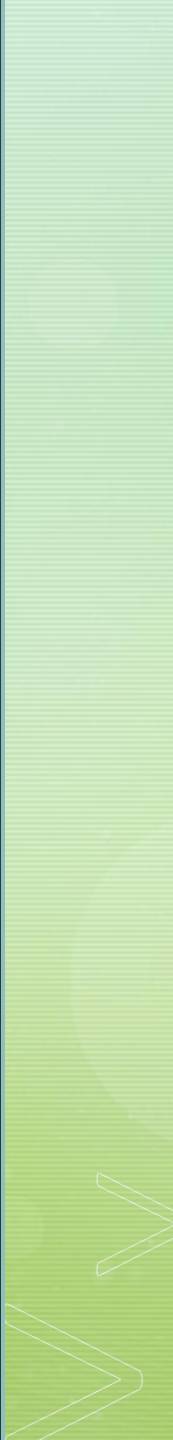


Core Ethical Principles

- Integrity of evidence
 - Confidentiality and privacy
 - Impartiality and objectivity
 - Accountability
- 

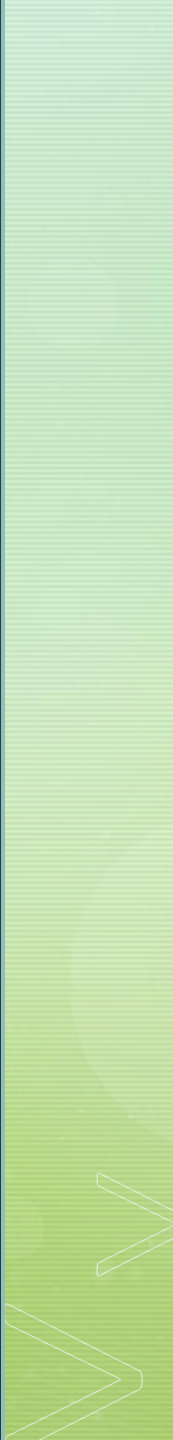


Legal Frameworks

- Computer Fraud and Abuse Act (CFAA)
 - GDPR for data privacy
 - Chain of custody requirements
- 



Common Ethical Dilemmas

- Accessing personal communications without consent
 - Handling privileged or confidential data
 - Using hacking tools during investigations
- 

Case Study 1: BTK Killer (2005)

- Metadata from Word document on floppy disk led to arrest
- Question: Was accessing deleted files ethical and legal?

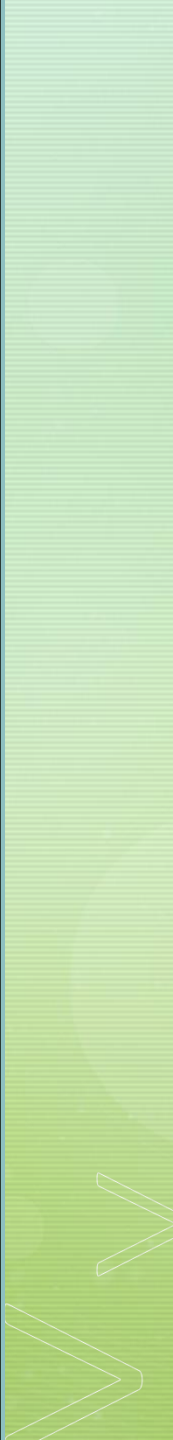


Case Study 2: Enron Email Investigation

- Massive email recovery for fraud evidence
- Issues: Employee privacy vs. corporate accountability

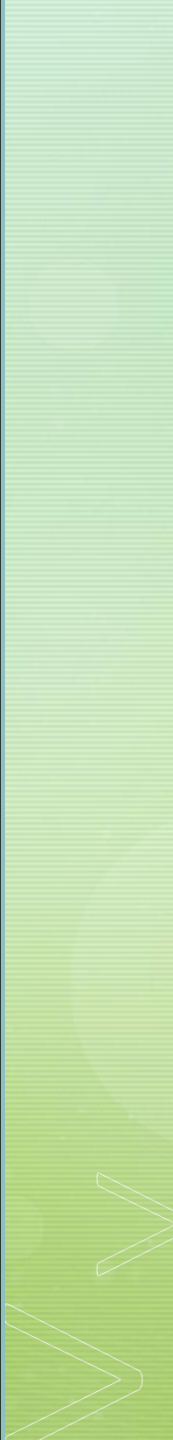


Case Study 3: iPhone San Bernardino (2016)

- FBI vs Apple encryption battle
 - Debate: National security vs user privacy
- 



Professional Codes of Conduct

- ISFCE Code of Ethics
 - ACM and IEEE ethical guidelines
 - Principle: 'Do no harm' and respect privacy
- 



Privacy vs Justice Debate

- Should investigators access all data for solving crimes?
- Where do we draw the line for personal privacy?

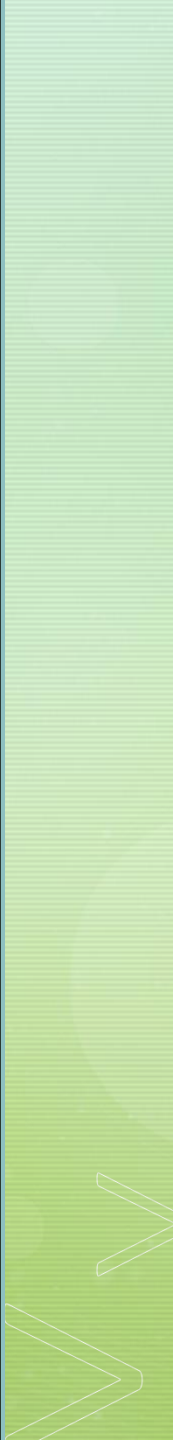


Activity: Class Debate

Divide into two groups:

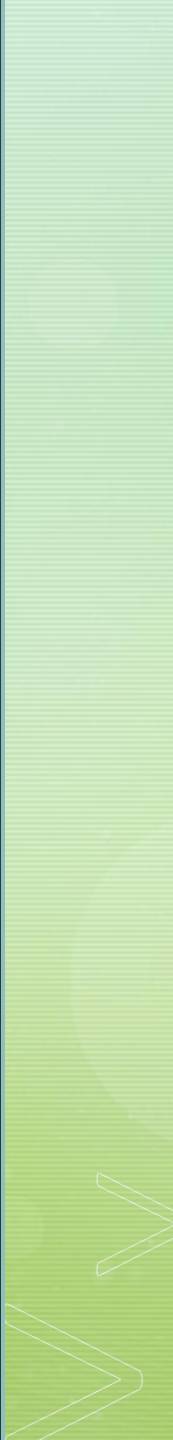
- Group A: Prioritize national security over privacy.
- Group B: Prioritize user privacy over government access.

“Prepare 3 key arguments on each”



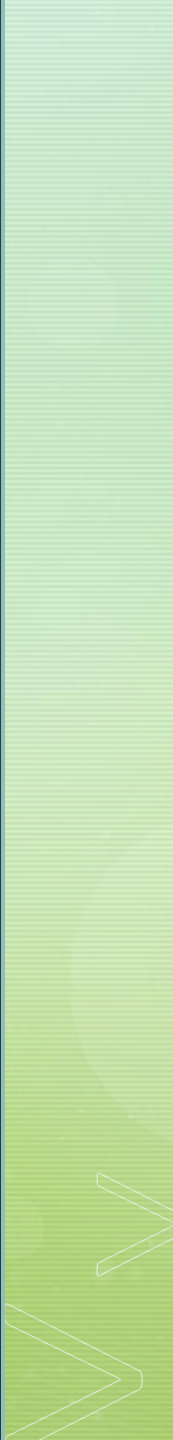


Challenges in Digital Forensics

- Encryption and anti-forensic tools
 - Cloud data jurisdiction issues
 - Big Data and AI complicating analysis
- 



Ethical Use of Forensic Tools

- Avoid planting evidence
 - Document every step
 - Use tools lawfully and transparently
- 

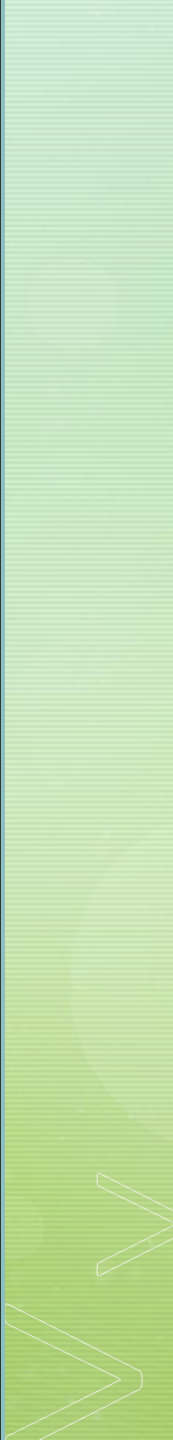


Handling Child Exploitation Cases

- Extreme psychological and ethical burden
- Need for controlled environments
- Mandatory reporting obligations



Corporate Investigations

- Employee monitoring ethics
 - Balancing IP protection with privacy
 - Avoiding misuse of investigation powers
- 



Scenario Analysis

Scenario: You find unrelated illegal content during a corporate investigation.

What steps should you take?

Discuss in pairs and justify your decision.

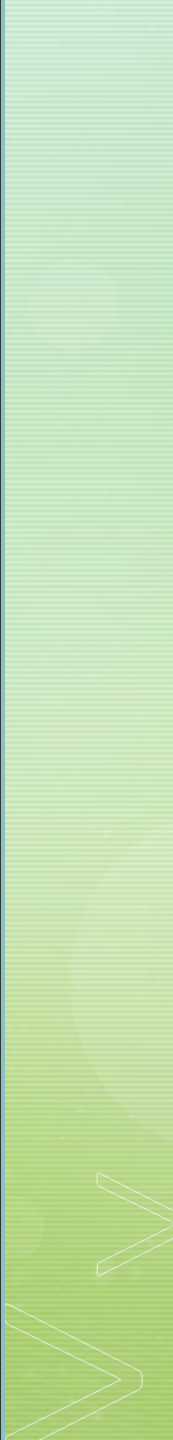


Emerging Trends & Ethical Risks

- Cloud and cross-border challenges
- AI-assisted forensics and bias
- Deepfake evidence manipulation



Best Practices

- Always maintain chain of custody
 - Seek warrants where required
 - Follow professional codes and training
- 

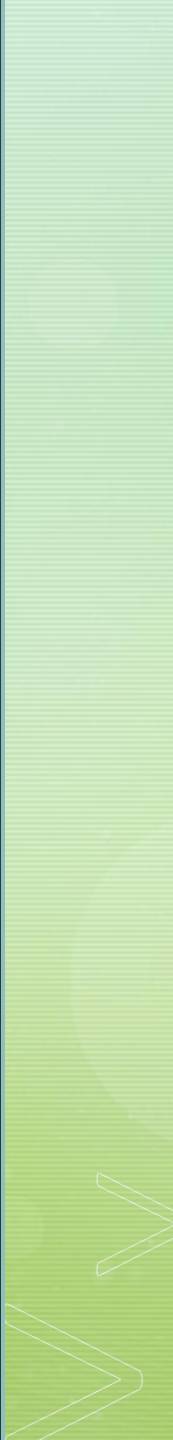


Quick Quiz

1. What is the chain of custody?
2. Name two professional ethics codes in forensics.
3. Give one ethical dilemma in mobile device forensics.



Summary & Takeaways

- Ethics ensures trust, legality, and justice
 - Respect privacy but uphold law
 - Critical thinking required in every case
- 



Ethics of Smart Cities and IoT

Engineering Ethics Module

Introduction to Smart Cities & IoT

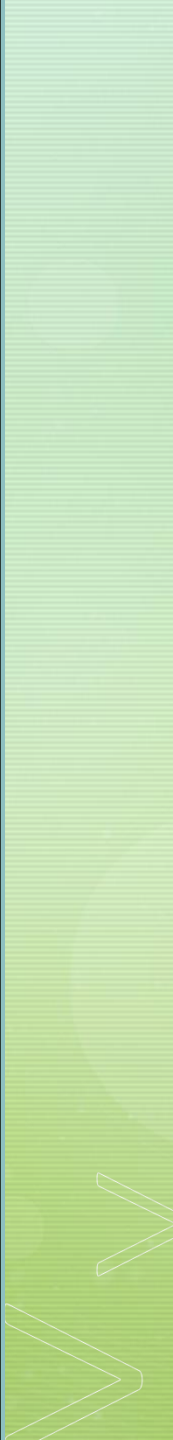
- Smart cities use IoT for data-driven services
- Applications: Traffic, energy, healthcare, surveillance
- Ethical implications are critical for engineers

Why Ethics Matters in Smart Cities

- Privacy concerns with pervasive data collection
- Algorithmic bias and fairness issues
- Security vulnerabilities impacting citizens

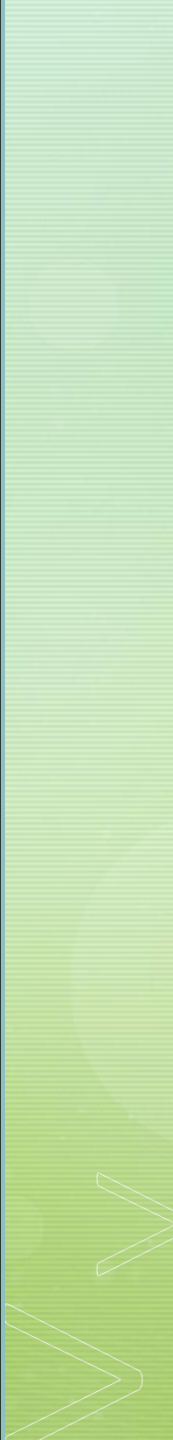


Core Ethical Principles

- Transparency in data collection
 - Accountability of decision-making systems
 - Respect for privacy and consent
 - Non-maleficence and fairness
- 

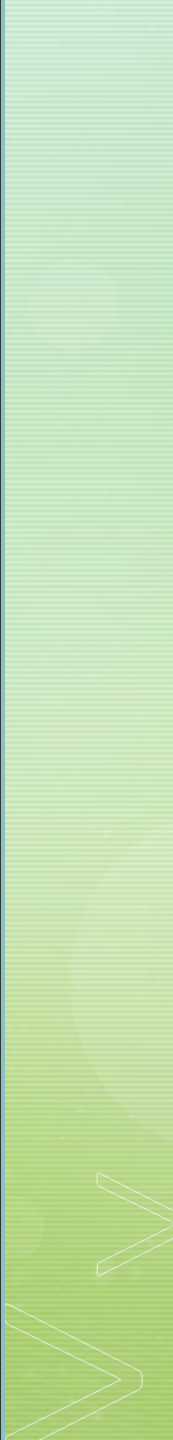


Key IoT Components

- Smart sensors and devices
 - Data analytics platforms
 - AI-driven decision engines
 - Cloud and edge infrastructure
- 



Ethical Challenges

- Mass surveillance vs public safety
 - Ownership and control of data
 - Algorithmic decision-making opacity
 - Cybersecurity threats
- 

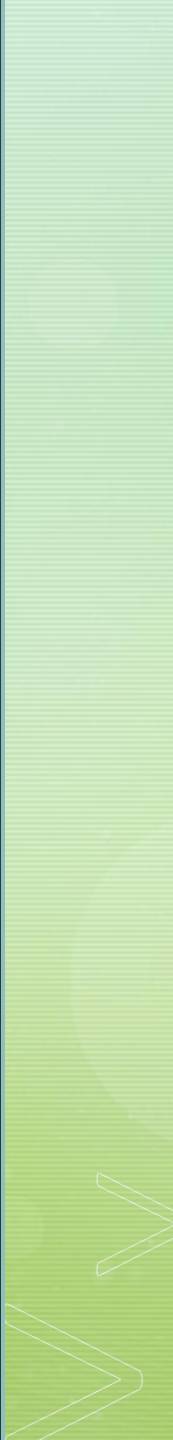


Legal & Policy Frameworks

- GDPR for data privacy
- Local data protection laws
- Smart city governance standards



Case Study 1: Sidewalk Labs (Toronto)

- Google's smart city project faced backlash over data privacy
 - Concerns: Who owns urban data? How is it monetized?
 - Result: Project canceled after ethical and governance concerns
- 



Case Study 2: Singapore Smart Nation

- Extensive IoT deployment for transport & security
- Debate: Security vs citizen privacy
- Ethical challenge: consent in public surveillance

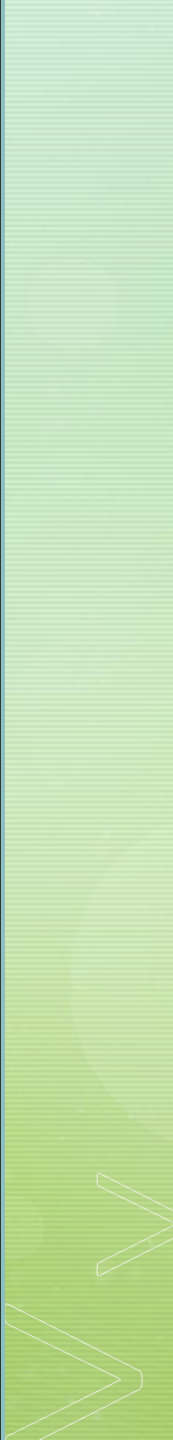


Case Study 3: Indian Smart Cities Mission

- Push for IoT and e-governance
- Challenges: Digital divide and data misuse
- Ethical issues: inclusion and data protection



Debate Prompt

- Should governments install facial recognition in public spaces?
 - Pros: Crime prevention, public safety
 - Cons: Privacy invasion, potential misuse
- 



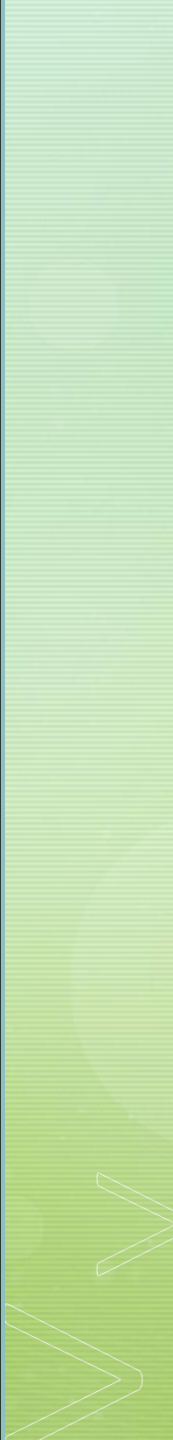
Activity: IoT Risk Analysis

Pick an IoT application (e.g., traffic sensors, smart meters).

Identify 3 potential ethical issues and propose mitigation strategies.

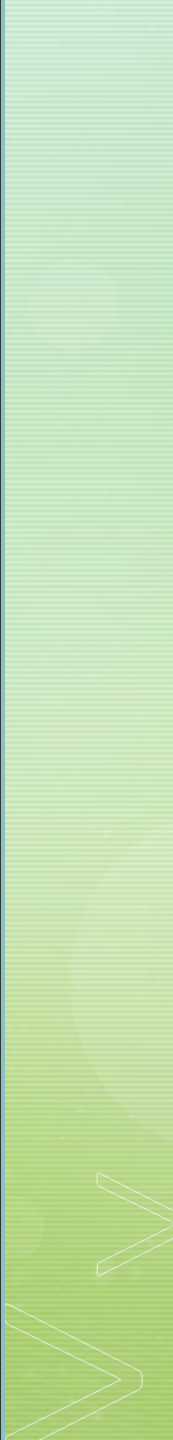


Security & Hacking Risks

- Smart grids hacked → power outages
 - Connected cars hijacked → safety threat
 - Data leaks from health IoT devices
- 



AI and Algorithmic Bias

- Predictive policing using biased data
 - Resource allocation algorithms discriminating against neighborhoods
 - Ethical duty: Ensure fairness and accountability
- 



Environmental & Sustainability Ethics

- Energy consumption of IoT devices
- E-waste and recycling concerns
- Green smart city initiatives

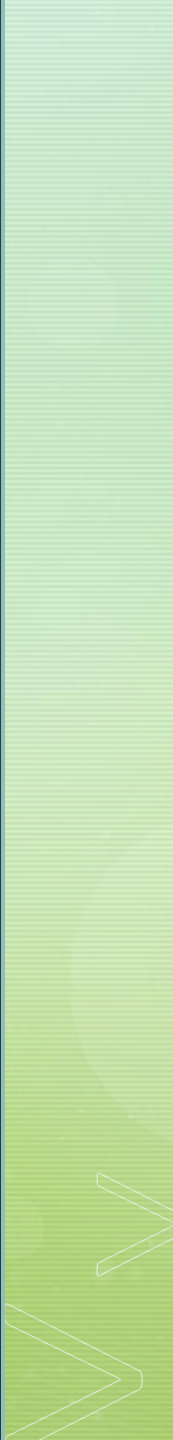


Professional Guidelines

- IEEE IoT ethics standards
- ACM Code of Ethics
- Principle: 'Respect privacy, prioritize safety'



Emerging Tech & Risks

- 5G and increased IoT connectivity
 - Digital twins and predictive models
 - Ethical dilemmas with autonomous systems
- 



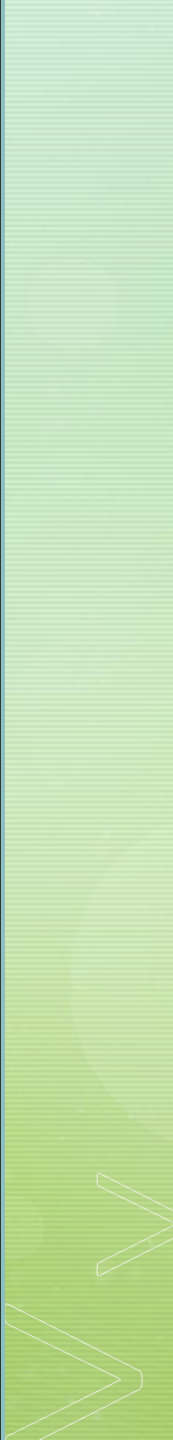
Scenario Analysis

Scenario: A smart traffic system prioritizes VIP routes, delaying ambulances.

Question on fairness and propose ethical fixes.

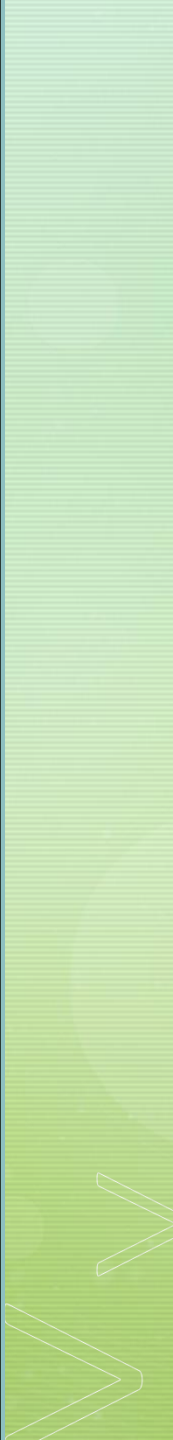


Best Practices

- Privacy by design
 - Secure IoT architecture
 - Transparency and stakeholder engagement
- 



Summary & Takeaways

- Smart cities promise efficiency but pose ethical risks
 - Balance innovation with rights and fairness
 - Engineers play a key role in ethical IoT design
- 

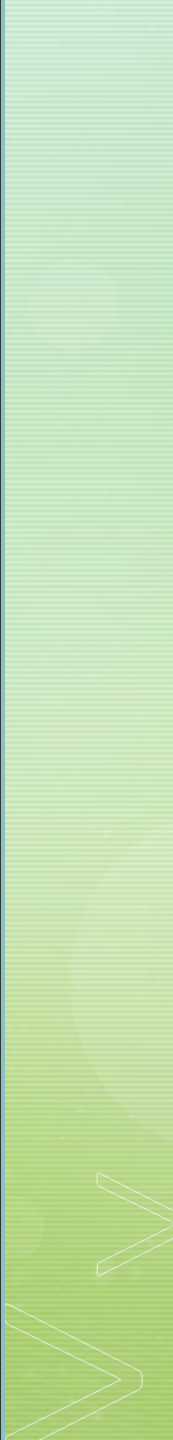


An Engineering Ethics Perspective

Quantum Computing and the Future of Cryptography: Ethical Risks

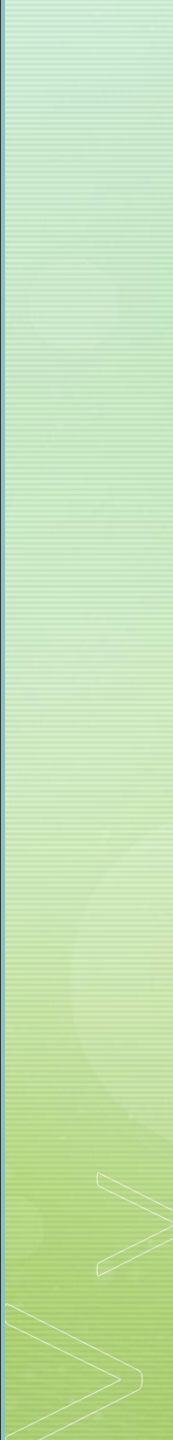


Introduction

- Quantum computing: a paradigm shift in computation.
 - Cryptography under threat from quantum advancements.
 - Focus on ethical risks and engineering responsibilities.
- 

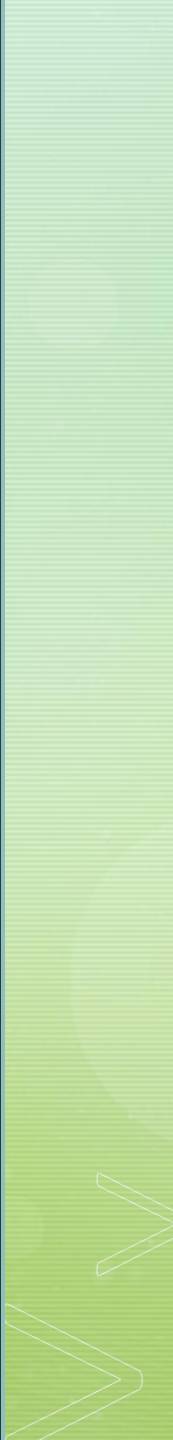


What is Quantum Computing?

- Uses quantum mechanics principles like superposition and entanglement.
 - Qubits replace classical bits.
 - Capable of solving complex problems exponentially faster.
- 

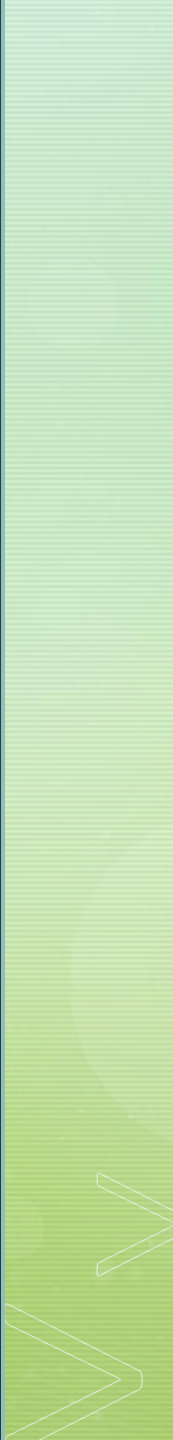


How Quantum Differs from Classical Computing

- Classical: 0 or 1 states; Quantum: superposition (0 and 1).
 - Entanglement enables faster problem-solving.
 - Potential to break current cryptographic methods.
- 



Introduction to Cryptography

- Science of securing communication and data.
 - Relies on mathematical algorithms and computational hardness.
 - Examples: RSA, ECC, AES.
- 

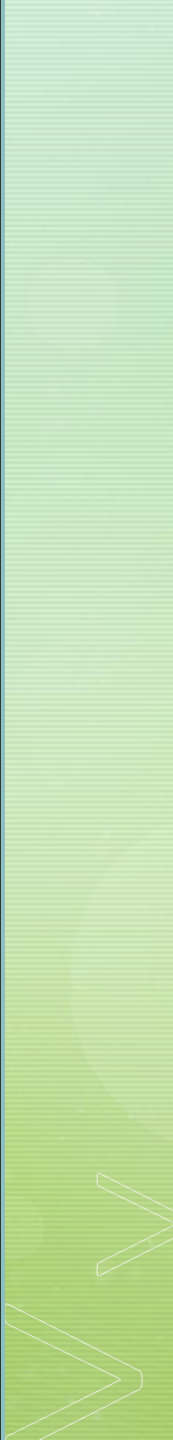


Quantum Threat to Cryptography

- Shor's algorithm can break RSA and ECC.
- Grover's algorithm weakens symmetric cryptography.
- Ethical risk: Massive data vulnerability.

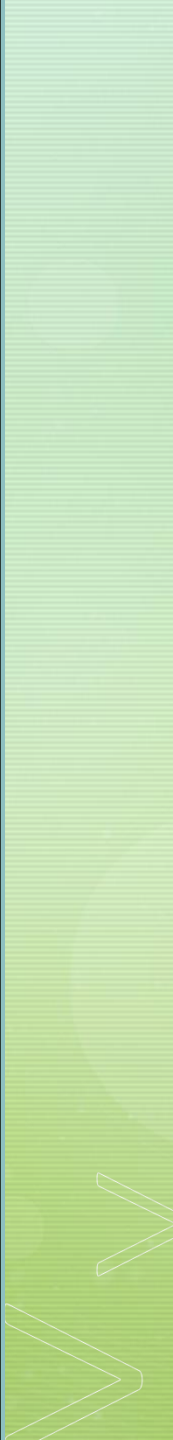


Ethical Dilemmas

- Should governments or companies deploy quantum cryptanalysis?
 - What about privacy rights of individuals?
 - Balance between national security and personal freedoms.
- 



Impact on Data Privacy

- Encrypted medical, financial, and personal data vulnerable.
 - Future risk: 'Harvest now, decrypt later' attacks.
 - Loss of trust in digital communication systems.
- 

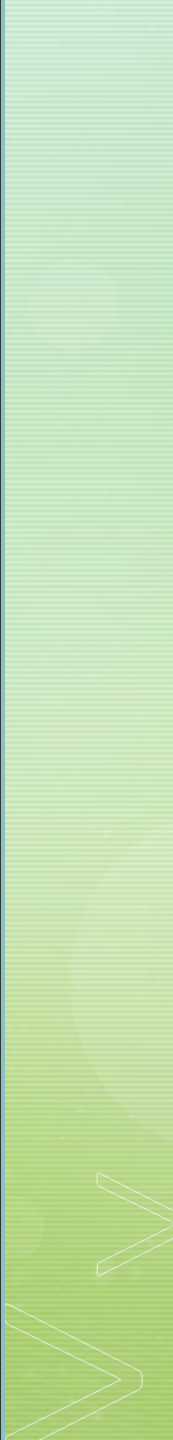


Impact on Cybersecurity

- Current infrastructures not quantum-safe.
- Critical sectors (banks, defense, healthcare) at risk.
- Raises ethical duty for engineers to prepare countermeasures.



Quantum-Safe Cryptography

- Post-quantum cryptography (PQC) being developed.
 - NIST standardization efforts ongoing.
 - Ethical responsibility: Transition planning.
- 

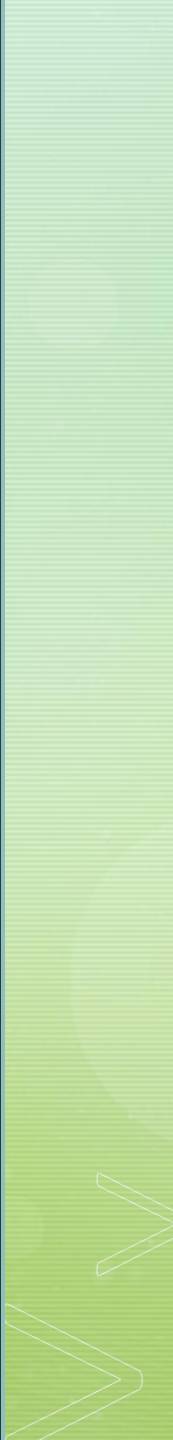


Blockchain and Quantum Risk

- Quantum can break blockchain signatures.
- Risk to cryptocurrencies and decentralized apps.
- Ethical concern: Loss of public trust in digital finance.



Military and Surveillance Risks

- Quantum code-breaking could give states massive surveillance powers.
 - Could destabilize global balance of power.
 - Ethical dilemma: Tech for peace vs. warfare.
- 

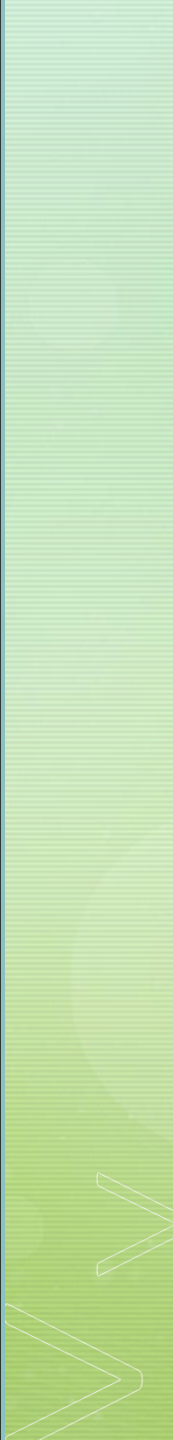


Economic and Business Implications

- Industries dependent on secure communication threatened.
 - Quantum readiness gap between companies.
 - Ethical obligation: Fair access to quantum security.
- 



Case Study: China vs. USA Quantum Race

- Geopolitical competition in quantum research.
 - Concerns about ethical misuse in surveillance.
 - Global standards urgently needed.
- 



Case Study: Banking and Finance

- Financial transactions heavily rely on encryption.
- Quantum could undermine trust in global banking systems.
- Urgent need for quantum-safe solutions.



Role of Engineers and Scientists

- Designing ethical quantum applications.
- Prevent misuse of quantum power.
- Advocate for responsible innovation.



Legal and Regulatory Challenges

- Lack of international quantum laws.
- Disparity in national approaches to regulation.
- Need for global ethical frameworks.



Ethical Frameworks for Quantum Computing

- Transparency in research and applications.
- Promote open collaboration for security solutions.
- Avoid monopolization of quantum resources.



Future of Quantum Cryptography

- Quantum Key Distribution (QKD) as a solution.
- Unhackable communication potential.
- Still ethical risks of inequality and misuse.



Balancing Innovation and Ethics

- Encourage beneficial quantum applications.
- Safeguard against misuse and harm.
- Engineers, policymakers, and society must collaborate.



Conclusion

- Quantum computing poses ethical risks to cryptography.
 - Urgent need for quantum-safe standards and cooperation.
 - Ethical responsibility to guide innovation safely.
- 